



Segmentation and Classification of Skin Cancer Diseases Based on Deep Learning: Challenges and Future Decisions

Abstract

This project proposes a dual-stage Deep Learning framework designed to enhance skin cancer detection by combining precise image segmentation with high-accuracy classification. By utilizing U-Net for boundary detection and architectures like EfficientNet or Vision Transformers (ViT) for feature extraction, the system aims to automate the screening of malignancies such as Melanoma and Basal Cell Carcinoma. However, the implementation faces significant real-world obstacles, including dataset bias toward lighter skin tones, class imbalances, and a lack of transparency in "black-box" neural networks. To bridge the gap between experimental success and clinical trust, the study advocates for a transition toward Explainable AI (XAI), the fusion of image data with patient metadata, and the optimization of lightweight models for mobile tele dermatology. Ultimately, while Deep Learning is a transformative tool for early diagnosis, its clinical integration depends on improving data diversity and algorithmic transparency.

Existing System

Traditional skin cancer diagnosis relies on the clinical ABCDE rule and dermoscopy, methods that are heavily dependent on a physician's subjective experience and prone to human error. While biopsy remains the definitive diagnostic standard, its invasive nature and slow turnaround time create a need for more efficient screening tools. However, early Computer-Aided Diagnosis (CAD) systems often struggle to fill this gap, as their reliance on manual feature extraction and hand-coded mathematics makes them ineffective when processing the complex, blurry, or low contrast images frequently encountered in real-world clinical settings.

Limitations of Existing System

Limitations are. The current diagnostic landscape is hampered by significant clinical and technical limitations. The subjectivity of manual inspection often leads to inconsistent diagnoses between clinicians, while a global shortage of specialists results in delayed detection that can allow malignancies to progress. From a technical standpoint, older automated systems suffer from low sensitivity and specificity, frequently failing to filter out "noise" such as body hair or inconsistent lighting. These inaccuracies contribute to a high false positive rate, which necessitates invasive biopsies that are often unnecessary, driving up medical costs and increasing patient anxiety.

Proposed System

The proposed system streamlines the diagnostic process through a dual-stage Deep Learning pipeline integrated into a user-friendly interface. In the first stage, a U-Net architecture performs pixel-wise segmentation to mask the lesion, effectively isolating it from healthy skin and common artifacts like hair or shadows. The second stage leverages transfer learning with pre-trained models such as MobileNetV2 or EfficientNet to categorize the isolated lesion into specific disease classes, including Melanoma and Basal Cell Carcinoma. This backend is paired



Matrusri Engineering College

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An Autonomous Institution | Approved by AICTE, Affiliated to Osmania University
16-1-486, Saidabad, Hyderabad-500059. Ph: 040-24072764



Department of Computer Science and Engineering

Email-ID: hodcse@matrusri.edu.in

Website: www.matrusri.edu.in

with a React/Next.js web dashboard, providing an intuitive platform for instant image uploads and feedback, which is optimized for the growing demands of tele dermatology.

Objectives

- To develop an automated, equitable, and interpretable Deep Learning framework for dermatological screening
- The framework enhances diagnostic accuracy,
- It reduces clinical workload
- ensures reliable performance across diverse patient populations through Explainable AI (XAI).

System Requirement Specifications:

a. Software Requirements:

- Programming: Python 3.9 +
- IDE: VS Code (with jupyter notebook) or pycharm
- Deep learning framework: TensorFlow 2.15 + and pytorch
- Open cv2 for image processing
- Numpy and Pandas for Scientific computing
- Sci-kit learn for ML utilities
- Matplotlib and Seaborn for Visualization

b. Hardware Requirements:

For Development:

- Personal computer or laptop
- CPU
- RAM: 8GB +
- GPU: RTX 3050 +
- Storage: 20GB +

Mini Project Batch No.:

Sl. No.	Roll No.	Name of the Student	Signature
1	1608-23-733-306	B V S S ADITYA	
2	1608-23-733-005	SAI KSHITISH S	
3	1608-23-733-045	SMARAN DESHMUKH	

Signature

Mrs. Bhagya Lakshmi
Designation